



Executive Summary

A finite-element thermal simulation was performed on a 12.0 mm × 12.0 mm semiconductor die fabricated from **Silicon**, discretized into a 30 × 30 computational mesh. The model incorporates 3 independent heat-generating functional blocks under an ambient boundary temperature of 25.0°C. The transient heat diffusion equation was solved over 100 time steps using an implicit (Backward Euler) scheme with convective boundary cooling.

Peak Junction Temperature	18.22 °C
Thermal Status	Within nominal operating range
Average Die Temperature	13.38 °C
Hotspot Coverage	0 mesh cells exceed 80°C

Chip Configuration

Material	Silicon
Chip Dimensions	12.0 mm × 12.0 mm
Mesh Resolution	30 × 30 elements
Ambient Temperature	25.0 °C
Solver Iterations	100
Thermal Conductivity	N/A W/m·K

Material Note: Standard semiconductor substrate material. Moderate thermal conductivity (148 W/m·K), industry default for ASIC dies.

Heat Source Configuration

Block Name	X-Range (mm)	Y-Range (mm)	Power Temp (°C)
CPU_Core_0	2.0 – 5.0	2.0 – 5.0	12.5
NPU_Accelerator	7.0 – 10.0	3.0 – 8.0	18.0
SRAM_Cache	1.5 – 5.0	7.0 – 10.5	4.0

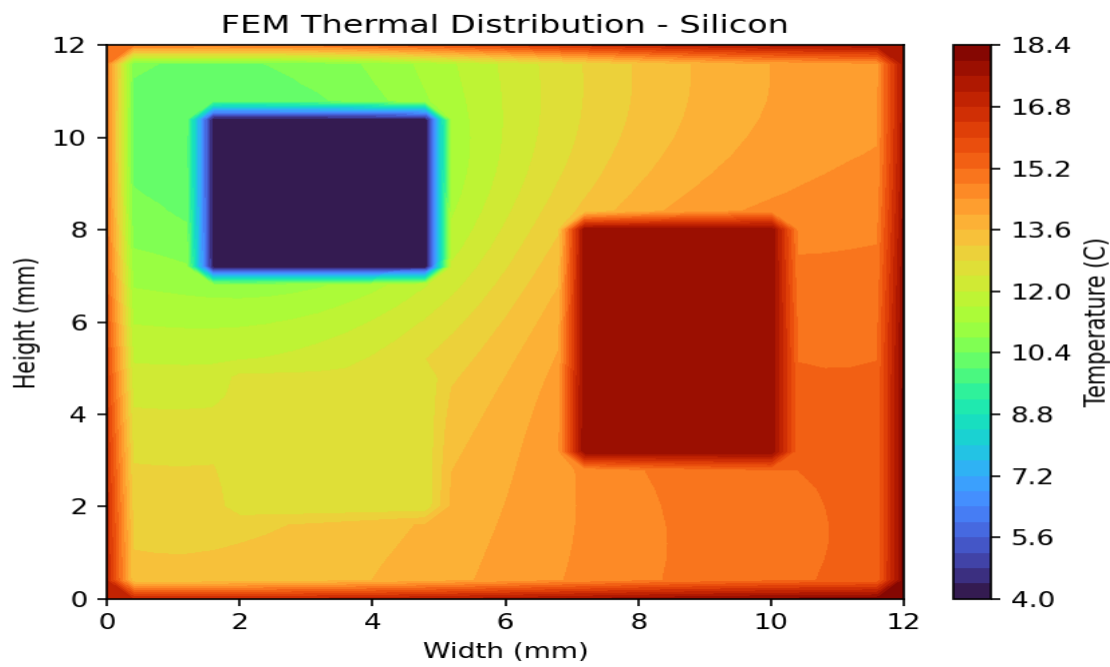
Hotspot Analysis Results

Maximum Temperature	18.22 °C
Average Temperature	13.38 °C
Temperature Variance	12.72
Hotspot Cells (> 80°C)	0

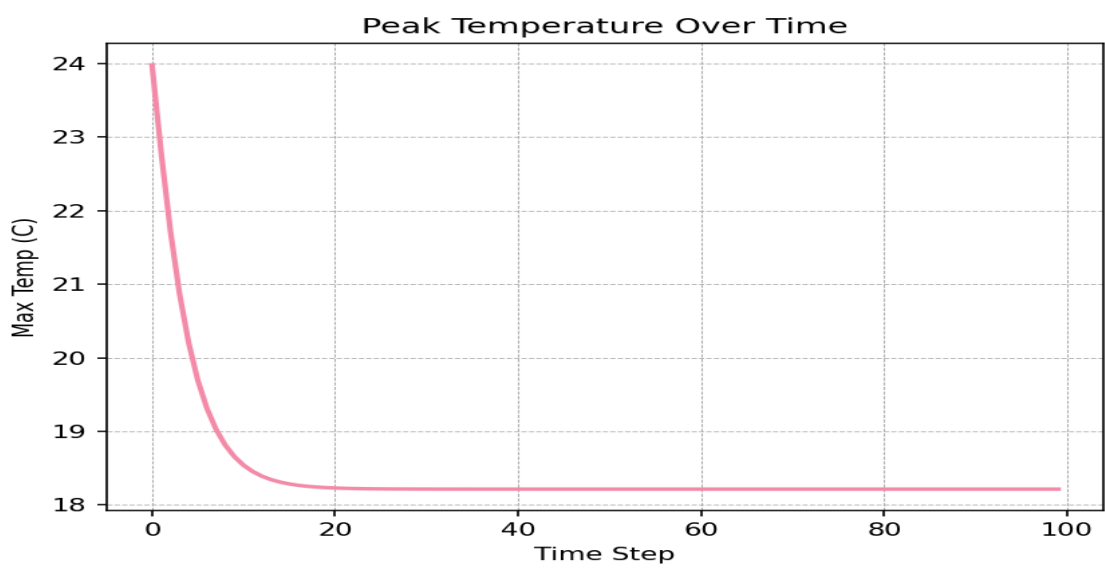
Status: Within nominal operating range



Thermal Distribution Map



Peak Temperature Over Time



Engineering Recommendations

- Peak temperature is within an acceptable range for the simulated configuration. Continued monitoring under sustained operating load is recommended.
- This simulation uses a simplified single-layer FEM model with convective boundary cooling. Results should be validated against physical prototype measurements before production sign-off.



Generated automatically by the Thermal Digital Twin engine. Results are simulation estimates derived from a finite-element approximation and should be validated against physical measurements before production sign-off.